Retail Performance Business Requirements

1 Business scenario

The retail management team needs one interactive report page to monitor sales, profitability, order activity, and product performance. Managers must be able to compare the current year with the equivalent period last year and identify the sports, product types, regions, and brands driving results.

The dashboard must support investigation of performance by state, sport, product colour, payment method, and promotion. It should help users identify where further investigation or commercial action is needed; it must not imply causes that the data does not establish.

2 Users and decisions

User	Decision supported
Retail leadership	Assess whether revenue and profit are improving against last year
Regional managers	Identify regional contribution and investigate state-level results
Merchandising team	Compare sports, brands, and product types for assortment review
Commercial analysts	Examine sales, margin, orders, and promotional segments together

3 Business questions

1. What are current YTD net sales and gross profit, and how do they compare with PYTD?
2. Is gross margin improving, and by how many percentage points?
3. How have units, order count, and average order value changed?
4. Which months show growth or decline against the same period last year?
5. Which sports generate the most YTD net sales?
6. Which product types combine strong sales with strong margins?
7. How much does each region contribute to YTD sales?
8. Which brands have the strongest and weakest YTD sales growth?
9. How do these answers change when a user applies the available filters?

4 Scope

Deliver one interactive Power BI report page containing a reporting-period header, six KPI cards, five analytical visuals, a left filter panel, and a reset button.

Included: sales and gross profitability, same-period comparisons, product and geographic segmentation, and report interactions.

Excluded: sales targets, forecasting, inventory availability, customer retention, market-share analysis, and claims about promotion effectiveness. These require additional data or analysis. Benchmark lines in this report are internal calculations, not market benchmarks.

5 Data requirements

Table	Role	Relevant fields
sales	Transaction fact table; verify one row per sales line	Sale_Line_ID, Order_ID, Purchase_Date, Product_ID, Store_ID, Net_Sales_INR, Net_COGS_INR, Gross_Profit_INR, Quantity, Returned_Qty, Payment_Method, Promotion
products	Product dimension	Product_Key, Product_Type, Sport, Brand_Label, Colour
store	Store dimension	Store_Key, State, Region
Date	Continuous calendar dimension	Date, Year, Month Num, Month Name, Month Short, Quarter

Create one-to-many relationships from unique product, store, and date keys to the matching fact columns. Verify matching key types, unmatched keys, and date-only relationships when Purchase_Date contains time. Use dimension-to-fact filtering unless a documented requirement justifies another design.

The Date table must include every day from 1 January 2025 through at least 15 September 2026 for the supplied exercise, and must extend when new source dates arrive. Sort month labels by Month Num. Do not invent a Category column; use the available Sport and Product_Type fields.

6 Reportingperiod rules

10. Determine the latest valid purchase date from the full loaded sales data. For the reference exercise, the intended cutoff is 15 September 2026; validate this against the dataset.
11. YTD covers 1 January of the cutoff year through that cutoff date, inclusive.
12. PYTD covers 1 January of the previous year through the corresponding prior-year cutoff, inclusive: 1 January–15 September 2025 in this example.
13. Keep the report cutoff fixed when business slicers change. Apply the same selected business segment to both current- and previous-year calculations.
14. The monthly chart shows individual monthly sales, not cumulative YTD sales. Compare full January–August months and 1–15 September in both years. Months after the cutoff are blank or hidden.
15. Display both years and the cutoff clearly in the header. Indicate that September is a partial month.
16. If future data includes a leap-day cutoff, use a documented prior-year equivalent, such as 28 February for a 29 February cutoff.

7 KPI requirements

Each card must show the primary YTD result, the PYTD result, and the year-over-year change.

KPI	Business definition	Comparison	Display
Net Sales YTD	Sum of Net_Sales_INR in the YTD window	(YTD – PYTD) / PYTD	INR, millions
Gross Profit YTD	Sum of validated Gross_Profit_INR in the YTD window	(YTD – PYTD) / PYTD	INR, millions
Gross Margin YTD %	YTD gross profit / YTD net sales	YTD margin – PYTD margin, in percentage points	Percentage; change in pp
Units Sold YTD	Sum of units using the agreed sold-unit definition	(YTD – PYTD) / PYTD	Thousands
Total Orders YTD	Distinct count of valid Order_ID in the YTD window	(YTD – PYTD) / PYTD	Thousands

KPI	Business definition	Comparison	Display
AOV YTD	YTD net sales / YTD distinct orders	(YTD − PYTD) / PYTD	INR per order

Calculation rules

- Validate whether Quantity already reflects returns. If it represents gross sold units, explicitly choose gross units or net units (Quantity minus Returned_Qty) and use the same definition for both years. Do not subtract returns twice.
- Validate that Net_Sales_INR already incorporates the intended discount and refund treatment; do not deduct these a second time.
- Reconcile gross profit to net sales minus net COGS according to the dataset definition.
- Calculate margin as the ratio of aggregated profit to aggregated sales, not the average of row-level margins.
- Calculate AOV from aggregated sales and distinct orders, not an average of line-level values.
- Confirm Order_ID uniqueness across stores/channels before using it as the business order key. Exclude invalid blank identifiers.
- A margin change from 40.9% to 41.2% is +0.3 percentage points, not +0.3% relative growth.
- If a previous-year denominator is zero or missing, show a blank or N/A change rather than an infinite or misleading percentage.
- Use unrounded values in calculations; round only for display.
- Keep numeric measures separate from text measures containing arrows and formatted values.
- Show positive changes with an upward arrow and green, negative changes with a downward arrow and red, and zero change neutrally.

8 Analytical visual requirements

Visual	Fields and measures	Business purpose and rules
Monthly Net Sales	Date month; CY monthly net sales; PY equivalent monthly net sales	Identify seasonality and year-over-year movement. Chronological Jan–Sep axis, common sales scale, solid blue CY and dashed muted PY, partial-period annotation.
Net Sales YTD by Sport	products[Sport]; Net Sales YTD	Rank sports descending. Show INR values. Average line is the arithmetic mean of YTD sales across the selected sports included in the comparison, with population explicitly defined.
Product Type Performance	One bubble per products[Product_Type]; X = Net Sales YTD; Y = Gross Margin YTD %; size = Total Orders YTD	Compare scale and profitability. Include product type, sales, profit, margin, and orders in tooltips. Use consistent bubble color to reduce clutter. Verify the existing bubble-size field before claiming it is orders.
Net Sales YTD by Region	store[Region]; Net Sales YTD	Show regional sales and contribution percentages. Center total must reconcile to the sales card. Shares must sum to approximately 100%, allowing rounding.
Sales Growth by Brand Label	products[Brand_Label]; Net Sales YoY % for YTD versus PYTD	Rank brands by growth. Show percentage labels and an explicitly named internal benchmark. Show negative growth correctly when present.

Benchmark definitions

- Scatter vertical line: arithmetic average of Net Sales YTD across the selected product types with sales data. Ignore the individual bubble context while retaining slicer selections.

- Scatter horizontal line: total selected YTD gross profit divided by total selected YTD net sales. This is a weighted overall margin, not a simple average of product-type percentages.
- Brand average line: if the label says Average brand growth, use the arithmetic average of valid brand YoY percentages in the defined selection. Exclude undefined growth from zero/missing prior-year sales.
- Alternative brand benchmark: use Overall sales growth, calculated from selected total YTD and PYTD sales. Label it explicitly. It need not equal Average brand growth.
- If a Top N filter is added, specify whether benchmarks describe all selected categories or only the displayed Top N. Keep the label and calculation consistent.

9 Filters and interactions

Slicer label	Field
State	store[State]
Sport	products[Sport]
Colour	products[Colour]
Payment Method	sales[Payment_Method]
Promotion	sales[Promotion]

Use dropdown slicers with clear headings and multi-select support. All intended report visuals must respond consistently to the selections. Keep current- and prior-year comparisons on the same selected business segment.

The reset button must return these five slicers to the documented baseline. Recommended baseline: all values included in every slicer. If using a bookmark, scope it to the intended slicers and capture their data state so unrelated visual visibility and page layout are not changed.

Do not confuse the literal Promotion value None with an unfiltered slicer. The screenshot shows Promotion = None while the other four slicers show All; consequently, its displayed KPI values may represent the no-promotion segment.

Chart selections should filter other visuals where the interaction supports analysis. Test KPI cards and benchmark lines after selection. Clearing a selection must restore the slicer-defined view.

10 Presentation requirements

- Title: Retail Performance. Subtitle: Sales & Profitability Overview.
- Place the filter panel on the left and all six KPI cards in a single horizontal row.
- Use a consistent blue accent (#3642B8), pale page background (#F4F7FB), white visual cards, and light borders (#E2E8F0).
- Highlight the Net Sales card using #C8CDF0; add thin blue top accents to KPI cards.
- Use readable titles, consistent INR units, aligned card values, and unclipped chart labels.
- Use colour plus arrows/text for changes; do not communicate meaning through colour alone.
- Display the user-friendly label Payment Method instead of Payment_Method.
- Keep subtitles clear about YTD and same-period comparisons. Avoid ambiguous titles when a visual uses a different date scope.
- Maintain sufficient plot area. If Top N is used to eliminate scrolling, state Top N in the title and keep the full data available through an appropriate interaction or separate detail view.

11 Screenshot observations to validate

These are review observations, not verified model defects.

17. The screenshot shows Net Sales YTD of approximately ₹39.82M while Promotion is set to None. Treat this as a reference filter state, not the required all-data total.
18. Visible Sport bars already sum to more than the YTD sales card. Check whether the chart uses all-period Net Sales, different filters, or an incorrect relationship. For this brief, the Sport chart must use the same YTD scope as the KPI.
19. Several positive YoY values are black while the sales change is green. Standardize their conditional formatting.
20. The brand reference line is labeled Avg. Define its population and calculation before teaching it as a benchmark.
21. The scatter bubble-size measure cannot be confirmed from the screenshot. This brief specifies orders; inspect the field well before documenting the existing implementation as compliant.

12 Acceptance tests

ID	Test	Expected result
AT01	Check calendar dates and dimension keys	No missing calendar dates, unique dimension keys, valid relationships
AT02	Validate cutoff and PYTD range	Matched cutoff periods; business slicers do not shift the global cutoff
AT03	Reconcile monthly CY values to YTD sales	January through cutoff-month values sum to the sales KPI under identical filters
AT04	Reconcile monthly PY values to PYTD sales	Equivalent prior-year months sum to PYTD KPI
AT05	Reconcile sales by sport and region	Each complete category breakdown sums to the sales KPI; account explicitly for blank/unmatched categories
AT06	Recalculate margin and AOV	Ratios of totals match card measures
AT07	Test duplicate order lines	Orders use distinct valid order keys, not transaction-line count
AT08	Test zero/missing prior-year sales	Growth shows blank/N/A without errors; benchmark excludes undefined values appropriately
AT09	Apply every slicer separately and in combination	All intended visuals and benchmarks respond consistently
AT10	Reset after multiple slicer changes	Documented baseline restored, including Promotion = All if that is the baseline
AT11	Inspect September comparison	Both years include only days 1–15 for the reference cutoff
AT12	Inspect final formatting	No truncated critical labels; readable units; consistent colours and arrows